

ICS Bayesian Risk Assessment Report

Executive Summary

This report presents the results of a quantitative Bayesian risk assessment for an Industrial Control System (ICS) environment. The assessment uses a Bayesian network model to compute compromise probabilities and risk scores based on the system topology, asset attributes, and any observed evidence.

Metric	Value
Overall Risk Score	0.719
Risk Level	Moderate
Assets Assessed	22
Connections Assessed	19
Evidence Used	None

Risk Register

Assets ranked by risk score (posterior probability × consequence impact). Higher scores indicate higher priority for investigation or treatment.

Rank	Asset	Risk Score	Probability	Risk Level
1	RTU	0.757	0.757	Moderate
2	Remote_Site	0.757	0.757	Moderate
3	PLC_2	0.757	0.757	Moderate
4	PLC_1	0.757	0.757	Moderate
5	HMI	0.757	0.757	Moderate
6	Safety_PLC	0.757	0.757	Moderate
7	OPC_Server	0.757	0.757	Moderate
8	Historian	0.757	0.757	Moderate
9	Domain_Controller	0.757	0.757	Moderate
10	Patch_Management_Server	0.757	0.757	Moderate
11	SCADA_Server	0.757	0.757	Moderate
12	Engineering_Workstation	0.757	0.757	Moderate
13	Corporate_Network	0.757	0.757	Moderate
14	VPN_Gateway	0.756	0.756	Moderate
15	DMZ	0.752	0.752	Moderate
16	Firewall	0.728	0.728	Moderate
17	Remote_Access_Server	0.609	0.609	Moderate
18	Internet	0.609	0.609	Moderate

Rank	Asset	Risk Score	Probability	Risk Level
19	Level_3_Switch	0.609	0.609	Moderate
20	Level_2_Switch	0.609	0.609	Moderate

Showing top 20 of 22 assets. Full register available in the CSV export.

Attack Path Analysis

The following attack paths represent the most likely routes an adversary could take through the network to reach high-value targets, ranked by combined propagation-and-target-risk score.

Path 1 — Score: 0.461

Internet → Firewall → DMZ → VPN_Gateway → Corporate_Network → SCADA_Server → OPC_Server → PLC_1 → RTU

Path 2 — Score: 0.461

Internet → Firewall → DMZ → VPN_Gateway → Corporate_Network → SCADA_Server → OPC_Server → PLC_1 → RTU → Remote_Site

Path 3 — Score: 0.461

Internet → Firewall → DMZ → VPN_Gateway → Corporate_Network → SCADA_Server → OPC_Server → PLC_1

Path 4 — Score: 0.461

Internet → Firewall → DMZ → VPN_Gateway → Corporate_Network → SCADA_Server → OPC_Server → PLC_2

Path 5 — Score: 0.461

Internet → Firewall → DMZ → VPN_Gateway → Corporate_Network → SCADA_Server → Historian

Methodology

This assessment uses a Bayesian network model constructed from the system topology. Each asset's base compromise probability is computed from its attributes (CVSS score, exposure, patch level). The Noisy-OR model generates conditional probability tables, and Variable Elimination performs inference given observed evidence. Risk scores combine posterior compromise probabilities with consequence impact scores.

Important: This report is a decision-support tool, not a guarantee of security. Risk scores are calculated approximations based on the configured model parameters and available evidence. They should be used alongside domain expertise and other security assessments.